

MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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Presentation Outline

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- Introduction
- Problem Statement & Objectives
- Scope of Project
- Project Applications
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Motivation

- Data remains underused in organizations.
- Analytics tools require technical expertise.
- Manual reporting is slow and inconsistent.
- LLM insights need verification.
- Agentic AI enables automated analysis.

Introduction

- Web-based multi-agent data analyst.
- Targets financial and sales CSV data.
- Uses LangGraph for stateful orchestration.
- Combines statistics and LLM reasoning.
- Produces validated reports and charts.

Problem Statement & Objectives

- Convert raw data into insights.
- Reduce manual cleaning effort.
- Generate professional analytical reports.
- Enable analytics without coding.
- Verify claims before reporting.

Scope of Project

- Web upload supports CSV datasets.
- Internal loader supports more formats.
- Six-agent pipeline is implemented.
- FastAPI backend provides services.
- React frontend supports workflow.

Project Applications

- Sales performance tracking.
- Financial reporting and budgeting.
- Retail and e-commerce analytics.
- Small business operations analysis.
- Reports for non-finance teams.

Methodology-(a) Qwen LLM Architecture

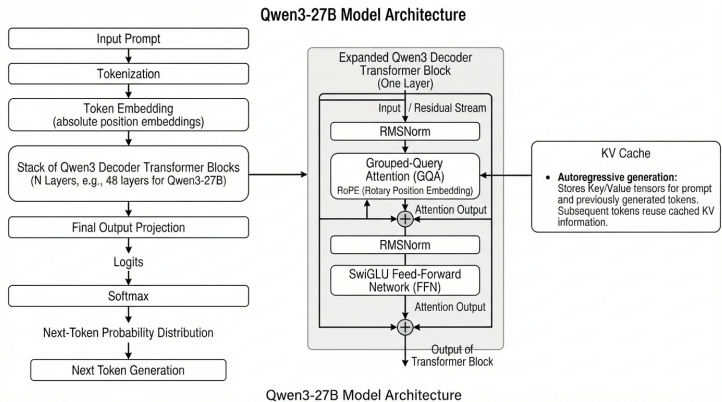


Figure: LLM Architecture

Methodology-(b) Backend Architecture

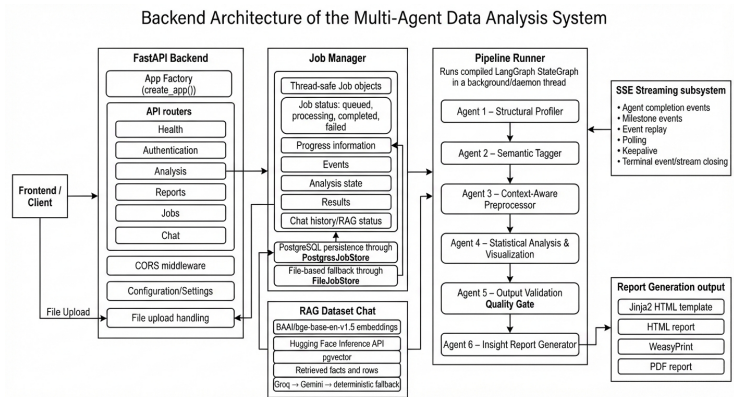


Figure: Backend Architecture

Methodology-(c) Frontend Workflow

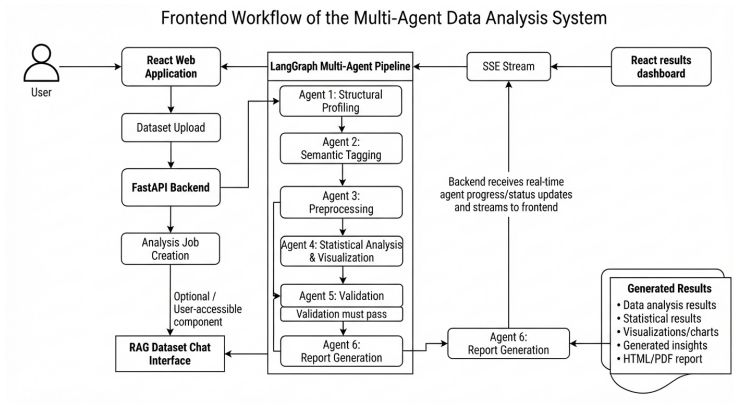


Figure: Frontend Flow

Methodology-(d) Agentic Workflow

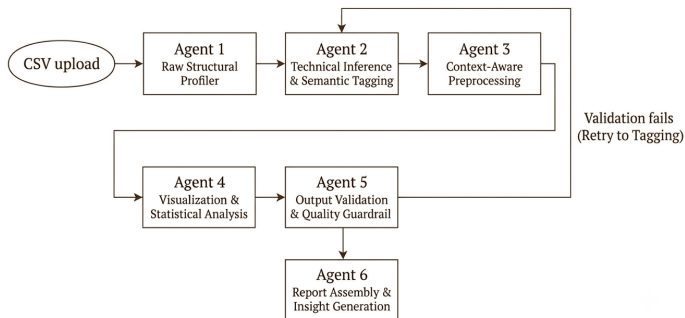


Figure: Agentic Workflow

Agent 1: Structural Profiler

- Loads the raw CSV file.
- Profiles shape and types.
- Detects missingness and duplicates.
- Caches DataFrame downstream.

```
[Agent 1] Profiled: 10000 rows x 21 cols | Missing: 0.0% | Duplicates: 0
```

Figure: Agent 1 Structural Profiler

Agent 2: Semantic Tagger

- Sniffs technical column types.
- Uses Groq with Gemini fallback.
- Builds processing blueprint.
- Adds confidence scores.

```
[Agent 2] Type sniffing summary: {'unknown': 16, 'numeric': 5}
[Agent 2] Blueprint built for 21 columns
[Agent 2] Semantic tags by column:

{
  order_id: {
    semantic_tag: identifier
    intended_type: string
    imputation: drop
    encoding: none
    identifier: True
    analysis_allowed: True
    confidence: 70.0
    null_action: drop_rows
    suitable: True
    reason: identifier_clean
  },
  order_date: {
    semantic_tag: datetime
    intended_type: datetime
    imputation: none
    encoding: none
    identifier: False
    analysis_allowed: True
    confidence: 60.0
    null_action: none
    suitable: True
    reason: datetime_missingness
  }
}
```

Figure: Agent 2 Semantic Tagger

Agent 3: Preprocessor

- Cleans currency and text.
- Coerces dates and numbers.
- Handles imputation and outliers.
- Derives business metrics.

```
[Agent 3] Preprocessing: profile=strict domain=finance_sales input=10000x21  
[Agent 3] Completed: rows=10000->10000 cols=21->115 quality=100.0/100 raw_missing=0.0% remaining_nulls=0.0%  
[Agent 3] Cleaning summary: deduped=0, scaled=1, added_cols=94, removed_rows=0  
[Agent 3] Note: the cleaned dataset is normalized and exported to outputs/cleaned_data.csv; if the row count stays the same  
rows to drop.  
[Agent 4] Starting analysis: 10000 rows x 115 cols
```

Figure: Agent 3 Preprocessor

Agent 4: Statistical Analysis

- Computes descriptive statistics.
- Detects correlations and anomalies.
- Scores chart families.
- Saves chart artifacts.

```
[Agent 4] Starting analysis: 10000 rows x 115 cols  
[Agent 4] Step 1 – Descriptive stats: 27 columns  
[Agent 4] Step 2 – Correlation done, strong pairs: 8  
[Agent 4] Step 3 – Growth rates skipped (no revenue/time axis)  
[Agent 4] Step 4 – Rankings skipped (no revenue/category pairing)  
[Agent 4] Step 5 – Seasonality skipped (no time axis)  
[Agent 4] Step 6 – Anomalies: 235 unique rows (2.35%) across 1 column  
[Agent 4] Step 7 – Distributions done (4 charts)  
[Agent 4] Step 8 – Regression done (27 columns, 0 charts)  
[Agent 4] Step 9 – Distribution charts done (1 charts)  
[Agent 4] Step 10 – Derived metrics skipped (no derived metrics)  
[Agent 4] Done – 6 charts saved to outputs/charts/
```

Figure: Agent 4 Statistical Analysis

Agent 5: Validation Gate

- Checks deterministic contracts.
- Validates chart integrity.
- Applies business rules.
- Blocks unsafe reports.

```
[Agent 4] Done: 10 charts saved to outputs/charts/  
[Agent 5] Starting output validation  
[Agent 5] Completed: score=100.0/100 passed=9/9 warned=0 failed=0 kappa=1.0
```

Figure: Agent 5 Validation Gate

Agent 6: Report Generator

- Extracts deterministic facts.
- Generates grounded narrative.
- Verifies numeric claims.
- Builds HTML/PDF report.

```
[Agent 6] Starting insight report generation
2026-08-25 17:16:42,771 [INFO] https: HTTP Request: POST https://api.groq.com/openai/v1/chat/completions "HTTP/1.1 200 OK"
[Agent 6] WARNING: 1 contradictory statement(s) detected in narrative. Review data quality claims for consistency.
2026-08-25 17:16:44,898 [INFO] weasyprint.progress: Step 1 - Fetching and parsing HTML - HTML string
2026-08-25 17:16:45,165 [INFO] weasyprint.progress: Step 2 - Fetching and parsing CSS - https://fonts.googleapis.com/css2?family=DM+Sans:ital,opsz,wght@0,9..40,400;0,9..40,600;0,9..40,700;1,9..40,400&family=SpaceGrotesk:wght@500;600;700&display=swap
2026-08-25 17:16:47,783 [INFO] weasyprint.progress: Step 2 - Fetching and parsing CSS - CSS string
2026-08-25 17:16:47,785 [WARNING] weasyprint: Ignored 'position: sticky' at 72:5, invalid value.
2026-08-25 17:16:47,795 [WARNING] weasyprint: Ignored 'background-filter: blur(10px)' at 73:5, unknown property.
2026-08-25 17:16:47,796 [WARNING] weasyprint: Ignored 'overflow-x: auto' at 81:38, unknown property.
2026-08-25 17:16:47,796 [WARNING] weasyprint: Invalid or unsupported selector: ':topnav::-webkit-scrollbar'
2026-08-25 17:16:47,797 [WARNING] weasyprint: Ignored 'position: sticky' at 101:14, invalid value.
2026-08-25 17:16:47,797 [WARNING] weasyprint: Ignored 'overflow-y: auto' at 101:94, unknown property.
2026-08-25 17:16:47,798 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 118:71, unknown property.
2026-08-25 17:16:47,799 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 129:25, unknown property.
2026-08-25 17:16:47,800 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 146:3, unknown property.
2026-08-25 17:16:47,801 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 153:103, unknown property.
2026-08-25 17:16:47,801 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 163:128, unknown property.
2026-08-25 17:16:47,802 [WARNING] weasyprint: Expected a media type, got '(max-width: 800px)'
2026-08-25 17:16:47,802 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 176:43, unknown property.
2026-08-25 17:16:47,803 [WARNING] weasyprint: Ignored 'box-shadow: inset 0 0 0 1px rgba(0,0,0,.08)' at 195:5, unknown property.
2026-08-25 17:16:47,803 [WARNING] weasyprint: Ignored 'box-shadow: inset 0 0 0 1px rgba(0,0,0,.08)' at 200:3, unknown property.
2026-08-25 17:16:47,805 [WARNING] weasyprint: Expected a media type, got '(max-width: 768px)'
2026-08-25 17:16:47,805 [WARNING] weasyprint: Invalid media type '(max-width: 768px)'
2026-08-25 17:16:47,805 [WARNING] weasyprint: Ignored 'box-shadow: var(--shadow)' at 235:5, unknown property.
2026-08-25 17:16:47,806 [WARNING] weasyprint: Expected a media type, got '(prefers-reduced-motion: reduce)'
2026-08-25 17:16:47,806 [WARNING] weasyprint: Invalid media type '(prefers-reduced-motion: reduce)'
2026-08-25 17:16:47,806 [WARNING] weasyprint: Expected a media type, got '(max-width: 1024px)'
2026-08-25 17:16:47,809 [WARNING] weasyprint: Invalid media type '(max-width: 1024px)'
2026-08-25 17:16:47,809 [WARNING] weasyprint: Ignored 'background-filter: none' at 315:33, unknown property.
2026-08-25 17:16:47,810 [WARNING] weasyprint: Ignored 'box-shadow: none' at 319:52, unknown property.
2026-08-25 17:16:47,811 [INFO] weasyprint.progress: Step 2 - Fetching and parsing CSS - CSS string
2026-08-25 17:16:47,814 [INFO] weasyprint.progress: Step 3 - Applying CSS
2026-08-25 17:16:47,852 [INFO] weasyprint.progress: Step 4 - Creating formatting structure
```

Figure: Agent 6 Report Generator

Methodology-(e) RAG Dataset Chat

- Indexes rows and facts.
- Uses BGE embeddings.
- Stores vectors in pgvector.
- Retrieves dataset context.
- Grounds answers in results.

Results-(a) Experimental Setup

- Dataset: 10000 Sales Records.csv.
- Input: 10000 rows, 14 columns.
- Profile: finance sales.
- Outputs: diagnostics and charts.
- Report generated from code.

Results-(b)Pipeline Execution Results

- Agent 1: no missing values.
- Agent 2: 14-column blueprint.
- Agent 3: 10000 x 58 output.
- Agent 4: 16 charts generated.
- Agent 5: 9/9 checks passed.
- Agent 6: 21/21 claims grounded.

Results-(c)Statistical Analysis Results

- 22 numeric columns summarized.
- 10 strong correlations found.
- 52 anomalous rows detected.
- Revenue/profit rankings generated.
- Seasonal charts generated.

Dashboard

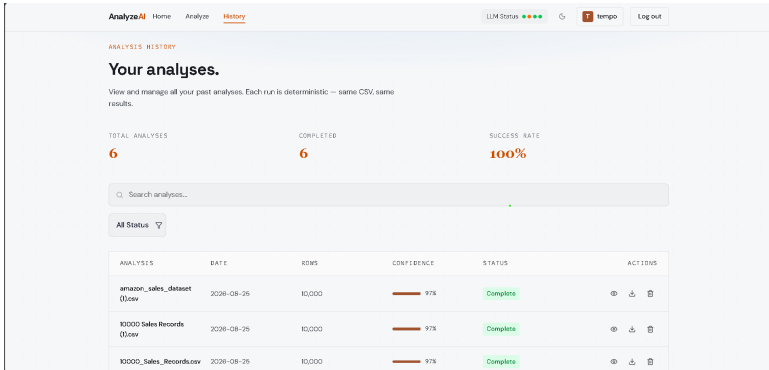


Figure: Dashboard

Final Generated Report

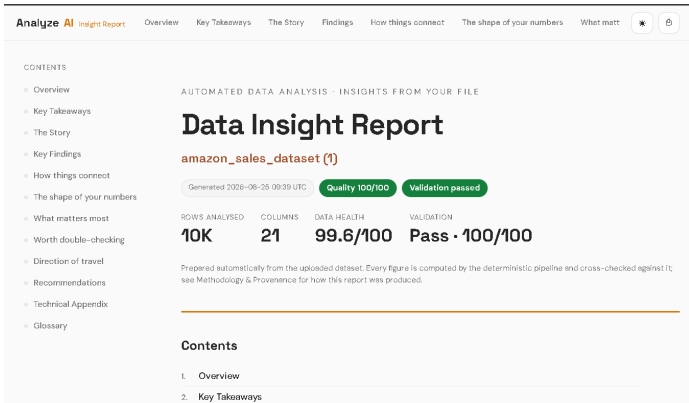
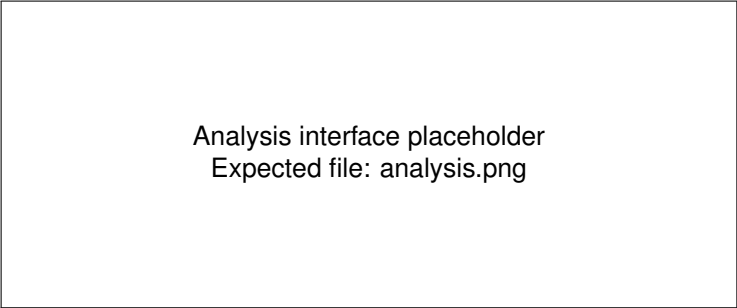


Figure: Final Generated Report

Analysis Interface



Analysis interface placeholder
Expected file: analysis.png

Figure: Analysis Interface

Analysis-RAG Chat Evaluation

- Retrieval evaluated at $K=8$.
- $\text{Precision@8} = 0.5938$.
- $\text{nDCG@8} = 0.6227$.
- $\text{MRR@8} = 0.7500$.
- Fallback answer path exists.

Analysis / Discussion I

- Pipeline completed in about 52s.
- Standalone Qwen took 4m 12s.
- Pipeline was about 4.8x faster.
- Qwen had scaling errors.
- Validation blocked unchecked claims.

Analysis / Discussion II

- Validation score was 100/100.
- Cohen's kappa was 1.0.
- Overall confidence was 0.97.
- Decision readiness was ready.
- Token metrics need instrumentation.

Future Enhancements

- Extend API file ingestion.
- Add provider configuration.
- Store API usage metrics.
- Improve chart interaction.
- Add feedback learning loop.

Conclusion

- Six-agent pipeline implemented.
- LangGraph controls execution.
- FastAPI and React provide access.
- Validation improves reporting trust.
- Project objectives achieved.

References I

1. Y. K. Dwivedi et al., “So what if ChatGPT wrote it?,” *International Journal of Information Management*, vol. 71, 2023.
2. OpenAI, “GPT-4 Technical Report,” 2024.
3. M. Sun et al., “LAMBDA: A Large Model Based Data Agent,” arXiv:2407.17535, 2024.
4. J. Maynez et al., “On Faithfulness and Factuality in Abstractive Summarization,” ACL, 2020.
5. V. Dibia and C. Demiralp, “Data2Vis,” *IEEE CG&A*, vol. 39, no. 5, 2019.

References II

6. K. Hu et al., “VizML,” CHI, 2019.
7. D. Moritz et al., “Formalizing Visualization Design Knowledge as Constraints,” *IEEE TVCG*, 2018.
8. Y. Luo et al., “DeepEye,” IEEE ICDE, 2018.
9. LangChain, “LangGraph: Stateful Multi-Agent Applications,” 2024.
10. Q. Wu et al., “AutoGen,” 2023.